5.4 Classwork: Introduction to Differential Equations (no calculator) — Markscheme

1. [Maximum mark: 9]

(a) $\frac{\mathrm{d}y}{\mathrm{d}x} = -2Ae^{-2x}$ **A1**

$6 - 2y = 6 - 2\left(Ae^{-2x} + 3\right)$ **M1**

$= -2Ae^{-2x}$ **A1**

so $\frac{\mathrm{d}y}{\mathrm{d}x} = 6 - 2y$ for every A **AG**

M1 for substituting y into the right-hand side. The final A1 needs both sides shown equal to $-2Ae^{-2x}$. Accept working from $\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 6$.

(b) $5 = A + 3$, so $A = 2$ **M1**

$y = 2e^{-2x} + 3$ **A1**

Wrong method: $5 = A$ (the $+3$ forgotten, since $e^0 = 1$) gives $y = 5e^{-2x} + 3$, M0A0.

(c) $y \to 3$ **A1**

When $y = 3$, $\frac{\mathrm{d}y}{\mathrm{d}x} = 6 - 2(3) = 0$ **A1**

Accept the argument $Ae^{-2x} \to 0$ for the first A1. The second A1 needs the substitution into the differential equation.

(d) $\dfrac{\mathrm{d}T}{\mathrm{d}t} = -k\left(T - 20\right)$ **A1A1**

A1 for $\frac{\mathrm{d}T}{\mathrm{d}t}$ proportional to $T - 20$. A1 for the negative sign (the temperature decreases while $T > 20$). $\frac{\mathrm{d}T}{\mathrm{d}t} = k(T - 20)$ with $k > 0$ earns A1A0. Accept $k\left(20 - T\right)$.

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2. [Maximum mark: 9]

(a) $\displaystyle\int \frac{1}{y}\,\mathrm{d}y = \int 6x^2\,\mathrm{d}x$ **M1**

$\ln|y| = 2x^3 + C$ **A1A1**

$\ln 2 = 0 + C$, so $C = \ln 2$ **M1**

$y = 2e^{2x^3}$ **A1**

M1 for separating with integral signs (or $\mathrm{d}y$ and $\mathrm{d}x$ on opposite sides). A1 for $\ln|y|$, A1 for $2x^3 + C$; the $+C$ must appear for the second A1. Accept $y = Ae^{2x^3}$ followed by $A = 2$. Wrong method: exponentiating and adding a constant afterwards, $y = e^{2x^3} + C$, gives $y = e^{2x^3} + 1$, earning M1A1A0M1A0.

(b) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = e^x e^{-y}$, so $\displaystyle\int e^y\,\mathrm{d}y = \int e^x\,\mathrm{d}x$ **M1**

$e^y = e^x + C$ **A1**

$e^{\ln 2} = e^0 + C$, so $2 = 1 + C$, $C = 1$ **M1**

$y = \ln(e^x + 1)$ **A1**

M1 for splitting e^{x-y} as $e^x e^{-y}$ and separating. Wrong method: taking logarithms term by term, $y = \ln e^x + \ln 1 = x$, earns the final A0.

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3. [Maximum mark: 9]

(a) $\displaystyle\int \frac{1}{T-20}\,\mathrm{d}T = \int -k\,\mathrm{d}t$ **M1**

$\ln|T-20| = -kt + C$ **A1**

$t = 0$, $T = 90$: $C = \ln 70$ **M1**

$T - 20 = 70e^{-kt}$ **A1**

$T = 20 + 70e^{-kt}$ **AG**

Accept $T - 20 = Ae^{-kt}$ with $A = 70$ for the second M1A1. The final A1 needs a line before the given answer.

(b) $55 = 20 + 70e^{-10k}$, so $e^{-10k} = \frac{1}{2}$ **M1**

$-10k = \ln\frac{1}{2} = -\ln 2$, so $k = \frac{\ln 2}{10}$ **A1**

AG

(c) $37.5 = 20 + 70e^{-kt}$, so $e^{-kt} = \frac{17.5}{70} = \frac{1}{4}$ **M1A1**

$kt = \ln 4 = 2\ln 2$, so $t = \frac{2\ln 2}{\ln 2/10} = 20$ minutes **A1**

Accept the half-life argument: the difference $T - 20$ halves every 10 minutes, $70 \to 35 \to 17.5$, so $t = 20$ (M1A1A1). Wrong method: linear cooling at 3.5 degrees per minute gives $t = 15$, M0A0A0.

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4. [Maximum mark: 10]

(a) $y_1 = 1 + 0.5(0 + 1) = 1.5$ **A1**

$y_2 = 1.5 + 0.5(0.5 + 1.5)$ **M1**

$= 2.5$ **A1**

M1 for a second step using the slope at $(0.5,\ 1.5)$, FT from the candidate's y_1. Wrong method: leaving out the step length, $y_1 = 1 + 1 = 2$, earns A0M0A0.

(b) $\frac{\mathrm{d}y}{\mathrm{d}x} - y = x$, integrating factor $e^{\int -1\,\mathrm{d}x} = e^{-x}$ **M1A1**

$\frac{\mathrm{d}}{\mathrm{d}x}(ye^{-x}) = xe^{-x}$

$ye^{-x} = \int xe^{-x}\,\mathrm{d}x = -xe^{-x} - e^{-x} + C$ **M1A1**

$x = 0$, $y = 1$: $1 = -1 + C$, $C = 2$, so $y = 2e^x - x - 1$ **A1**

First M1 for writing the equation in the form $y' + P(x)y = Q(x)$ with $P = -1$. Second M1 for integration by parts. An integrating factor of e^x (sign error) earns M1A0 and no further marks except FT of the constant.

(c) $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 1 + \frac{\mathrm{d}y}{\mathrm{d}x} = 1 + x + y$ **A1**

For $0 \le x \le 1$, $x \ge 0$ and $y > 0$, so $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} > 0$: the curve is concave up, so each Euler step along a tangent line lies below the curve. **R1**

Accept a comparison with the exact value $2e - 2$ for the R1 only if it is justified without a calculator, e.g. $e > 2.7$ gives $2e - 2 > 3.4 > 2.5$. The R1 needs a reason (concavity or the comparison), not just "Euler's method is approximate".

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5. [Maximum mark: 8]

(a) $y = vx$ gives $\frac{\mathrm{d}y}{\mathrm{d}x} = v + x\frac{\mathrm{d}v}{\mathrm{d}x}$ **A1**

$v + x\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{x^2 + v^2x^2}{x \cdot vx} = \frac{1 + v^2}{v}$ **M1**

$x\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{1 + v^2}{v} - v = \frac{1}{v}$ **A1**

AG

(b) $\int v\,\mathrm{d}v = \int \frac{1}{x}\,\mathrm{d}x$ **M1**

$\frac{1}{2}v^2 = \ln x + C$ **A1**

$x = e$, $v = \frac{2e}{e} = 2$: $2 = 1 + C$, $C = 1$ **A1**

$\frac{y^2}{2x^2} = \ln x + 1$, so $y^2 = x^2(2\ln x + 2)$ **M1**

$y = x\sqrt{2\ln x + 2}$ **A1**

Second M1 for substituting $v = \frac{y}{x}$ back. For the final A1 the positive root is required ($y > 0$). Accept $y = \sqrt{2x^2 \ln x + 2x^2}$. Wrong method: using $v = 2e$ (the value of y) in the initial condition gives $C = 2e^2 - 1$, earning A0 for the constant and A0 for the final answer.